



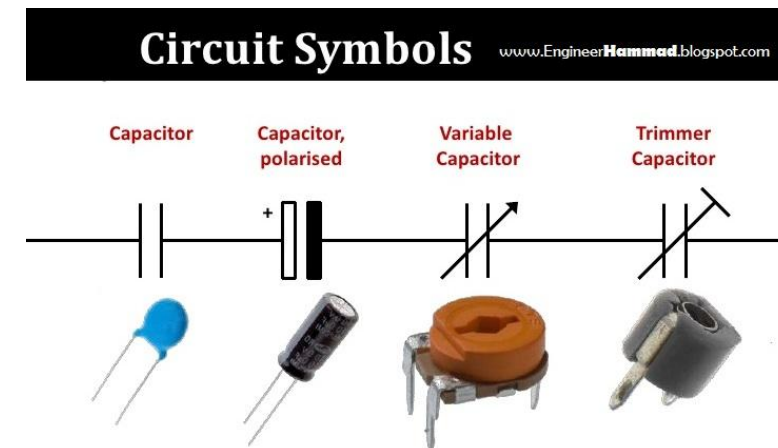
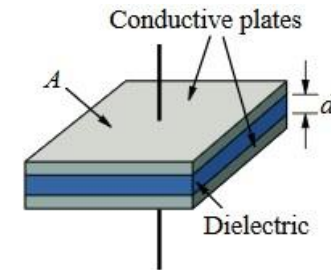
Introduction

Electrical & Electronic Circuits and Elements

Embedded Systems & Embedded Programming

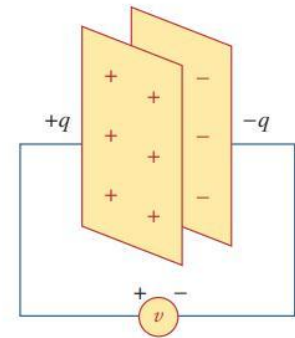
Fundamentals of Electric Circuits (Capacitor)

- ❖ Is a **passive element** which can **store energy** in its **electric field**
- ❖ Consists of two conducting **plates** separated by a **dielectric** (insulator)
- ❖ Air, ceramic, electrolyte etc. are used as dielectric
- ❖ Application
 - Capacitors in Series
 - Capacitors in Parallel
 - Coupling and Decoupling Capacitors
 - Energy Storage
- ❖ Types
 - Fixed Capacitors (the conducting surfaces are not adjustable)
 - Variable Capacitors (their capacitance can be changed mechanically or electronically, e.g. keypad)



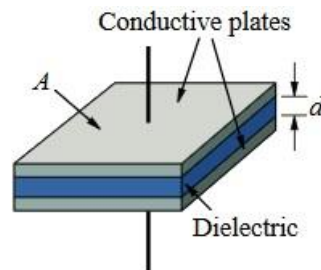
Fundamentals of Electric Circuits (Capacitor)

- ❖ By connecting to a voltage source it stores charge on the plates
- ❖ The amount of stored charge (q) is directly proportional to the applied voltage (v)
- ❖ Capacitance (C)
 - The amount of stored charge per plate for a unit voltage difference in a capacitor
 - More capacitance means more capacity to store charge
 - Its unit is **farads** (F) and can be measured by a **capacitance meter**
 - Depends on the permittivity of the dielectric and the physical dimensions of the capacitor



$$C = \epsilon A/d$$

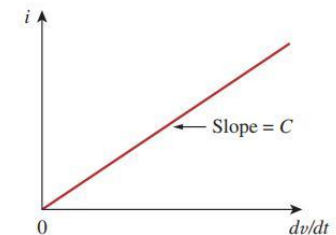
C : capacitance (F)
 ϵ : permittivity of the dielectric
 A : plate area
 d : distance between the plates



$$q \propto V \longrightarrow q = CV$$

$$\frac{dq}{dt} = C \frac{dv}{dt} \longrightarrow I(t) = C \frac{dv}{dt}$$

$$W_C(t) = \frac{1}{2} C v_C^2(t) \quad \text{Energy stored in a capacitor (J)}$$



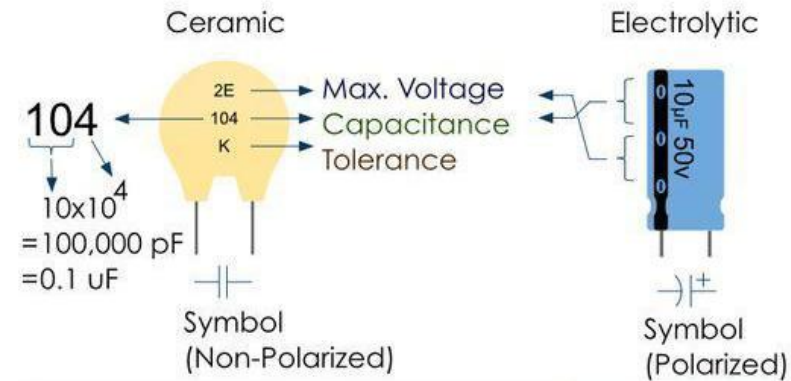
Current-voltage relationship of a capacitor.

Fundamentals of Electric Circuits (Capacitor)

- ❖ Capacitance (C) can be measured by
 - A capacitance meter
 - [Online calculator](#)
 - A cheat sheet



Capacitor Cheat Sheet



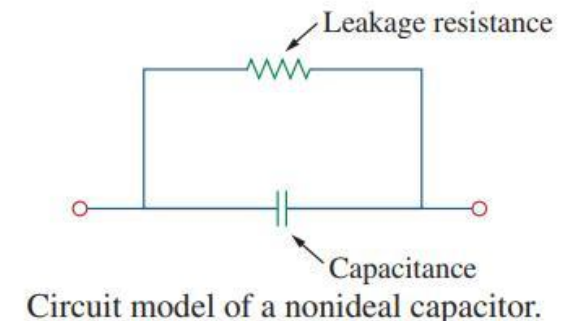
Capacitance Conversion Values		
Microfarads (µF)	Nanofarads (nF)	Picofarads (pF)
0.000001 µF	↔ 0.001 nF	↔ 1 pF
0.00001 µF	↔ 0.01 nF	↔ 10 pF
0.0001 µF	↔ 0.1 nF	↔ 100 pF
0.001 µF	↔ 1 nF	↔ 1,000 pF
0.01 µF	↔ 10 nF	↔ 10,000 pF
0.1 µF	↔ 100 nF	↔ 100,000 pF
1 µF	↔ 1,000 nF	↔ 1,000,000 pF
10 µF	↔ 10,000 nF	↔ 10,000,000 pF
100 µF	↔ 100,000 nF	↔ 100,000,000 pF

Tolerance	
Code	Percentage
B	± 0.1 pF
C	± 0.25 pF
D	± 0.5 pF
F	± 1%
G	± 2%
H	± 3%
J	± 5%
K	± 10%
M	± 20%
Z	+80%, -20%

Max. Voltage	
Code	Max. Voltage
1H	50V
2A	100V
2T	150V
2D	200V
2E	250V
2G	400V
2J	630V

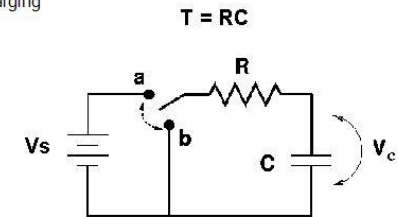
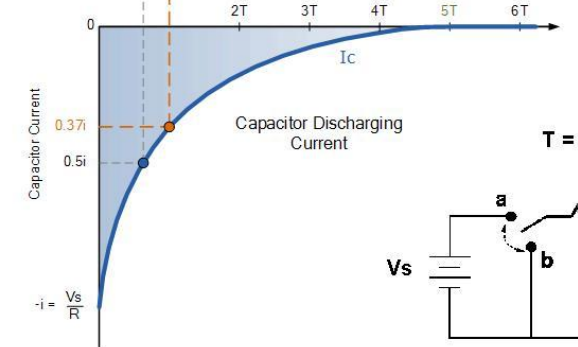
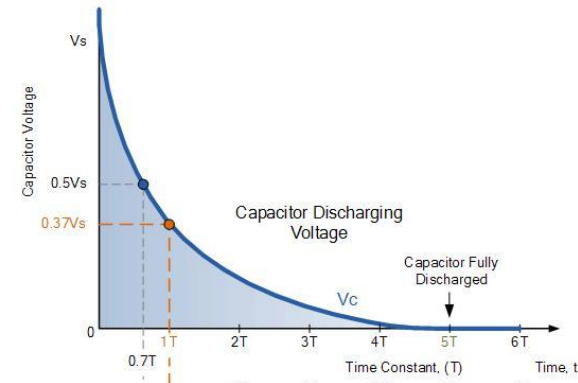
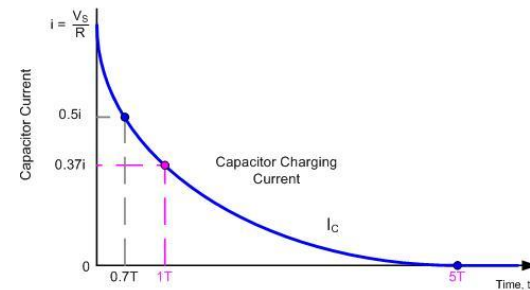
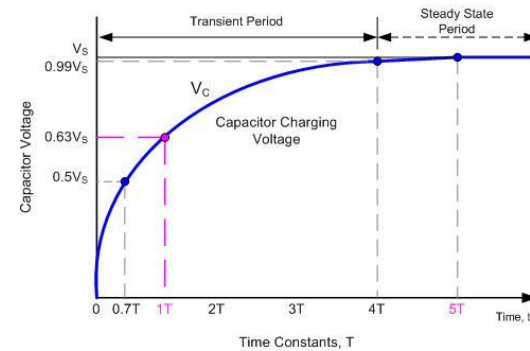
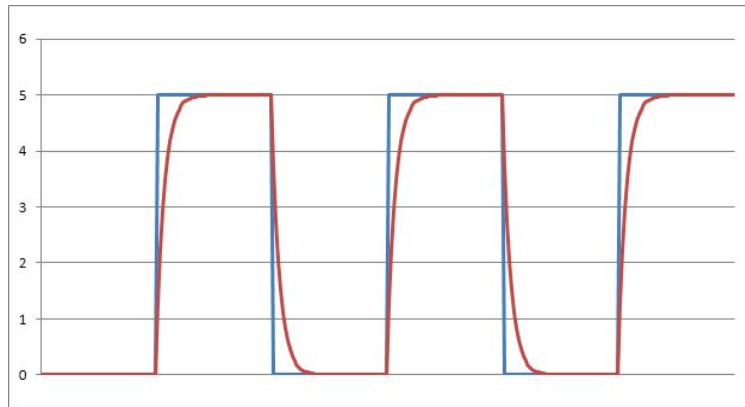
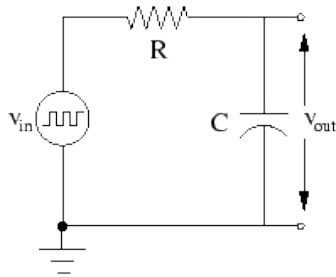
Fundamentals of Electric Circuits (Capacitor)

- ❖ A capacitor is an open circuit to dc voltage ($i = 0$)
- ❖ The voltage on a capacitor cannot change suddenly
 - Discontinuous change in the voltage requires an infinite current
- ❖ The ideal capacitor does not consume energy
 - Takes power from the circuit when storing energy (charging)
 - Returns previously stored energy when delivering power to the circuit (discharging)
- ❖ A real capacitor has a parallel-model leakage resistance ($100\text{ M}\Omega$)



Fundamentals of Electric Circuits (Capacitor)

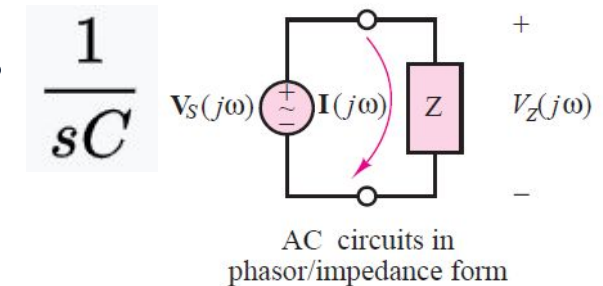
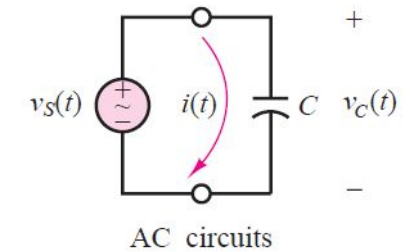
- ❖ Charge and discharge
- ❖ AC, puls and noise responses



[Capacitors and Capacitance](#)

Fundamentals of Electric Circuits (Capacitor)

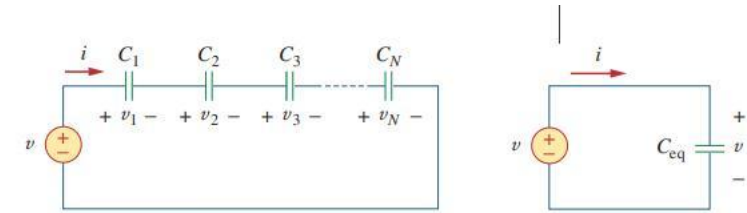
- ❖ Impedance (or complex resistance)
 - Is the **frequency-dependent** resistance of a load
 - Is the generalized version of the ohm's law
 - Is the i-v relationship of a load
 - Is measured in ohm(Ω)
 - Is represented by **Z** ($Z = R + jX$, X is the reactance)
- ❖ Impedance of a resistor is **R** and impedance of a capacitor is
 - Where **s** is the complex frequency of the source signal
 - **s** for a **DC** signal is **0** and for a **sinusoidal AC** signal is **$j\omega$**
 - **$\omega = 2\pi f$** where **f** is the frequency of the signal



Fundamentals of Electric Circuits (Capacitor)

❖ Capacitors in Series

- Capacitive voltage divider
- Filter out the DC component in a signal (Coupling capacitors)



$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots + \frac{1}{C_N}$$

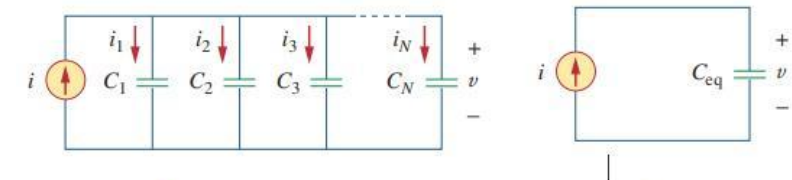
$$Q_{total} = Q_1 = Q_2 = \dots = Q_n$$

❖ Capacitors in Parallel

- DC power supplies
- Filter out the AC component (decoupling/bypass capacitors)

❖ Capacitors used for energy storage

- Audio equipment, UPS, camera flash etc.



$$C_{eq} = C_1 + C_2 + C_3 + \dots + C_N$$

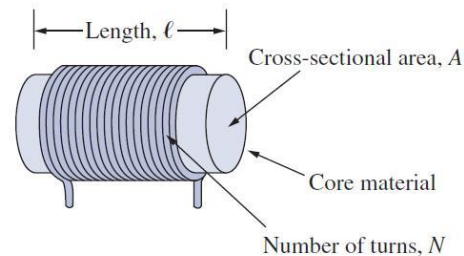
$$V_1 = V_2 = \dots = V_n$$

$$Q = Q_1 + Q_2 + \dots + Q_n$$

[Capacitor Guide](#)

Fundamentals of Electric Circuits (Inductor)

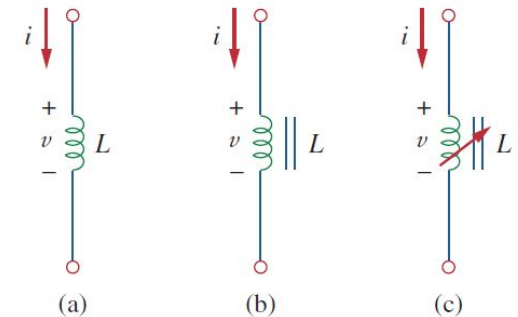
- ❖ Is a passive element that **stores energy** in its **magnetic field**
- ❖ Is used in power supplies, transformers, radios, radars, electric motors and etc.
- ❖ Any conductor has an inductive property and may be considered as an inductor
- ❖ To enhance the inductive effect, a practical inductor consists of a **coil of conducting wire**
- ❖ **Inductance** is the property that an inductor shows opposition to the change of current
 - Measured in henrys (**H**)
 - Depends on the physical dimension and construction of the inductor
 - Represented by **L**



Typical form of an inductor.

$$L = \frac{N^2 \mu A}{\ell}$$

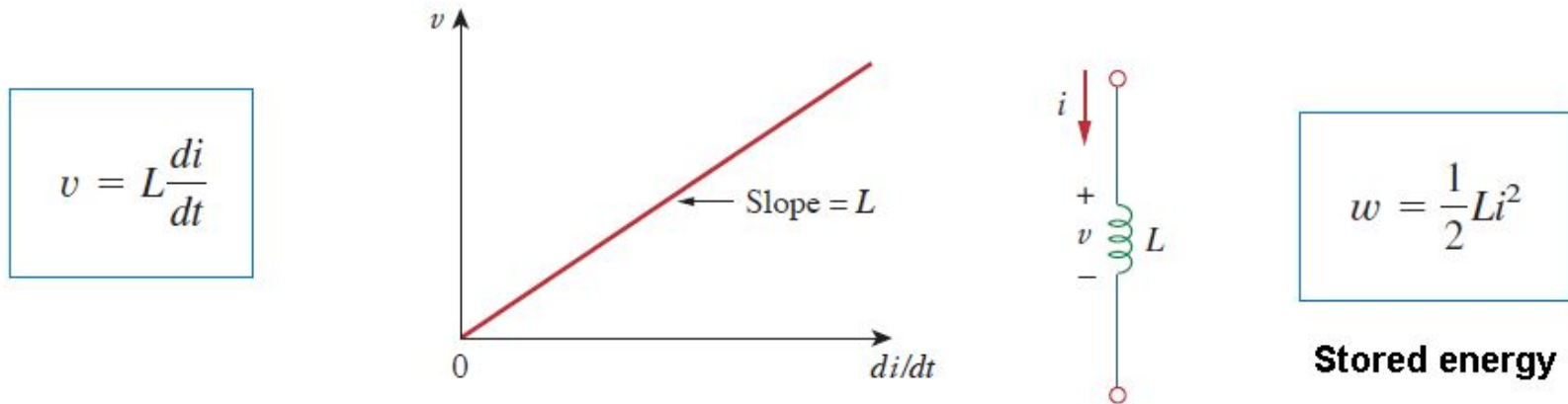
μ is the permeability of the core
 μ for air is 1.257×10^{-6} H/m



Circuit symbols for inductors: (a) air-core, (b) iron-core, (c) variable iron-core.

Fundamentals of Electric Circuits (Inductor)

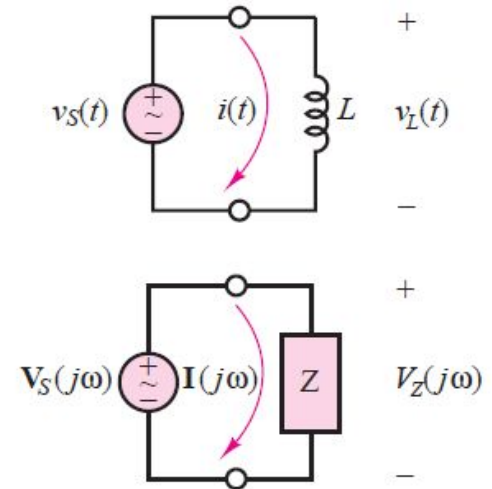
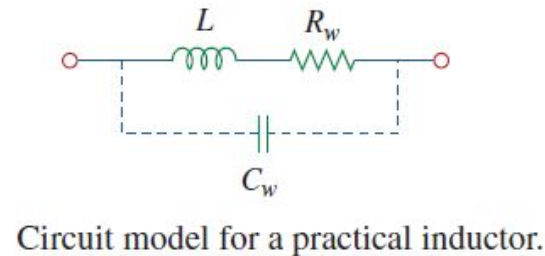
- ❖ The voltage across an inductor is directly proportional to the time rate of change of the current
- ❖ An inductor acts like a short circuit to dc current
 - The voltage across an inductor is zero when the current is constant
- ❖ The current through an inductor cannot change instantaneously
 - Discontinuous change in the current requires an infinite voltage



Voltage-current relationship of an inductor.

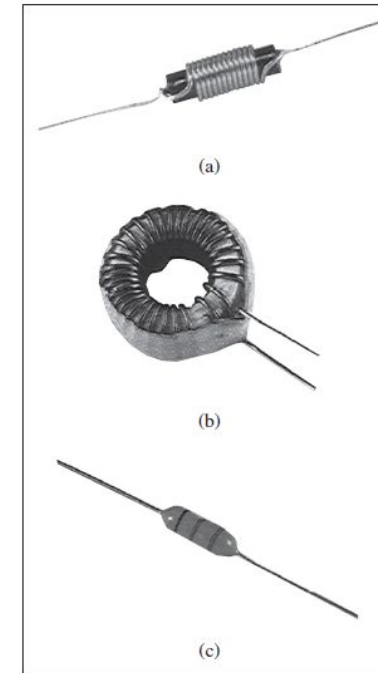
Fundamentals of Electric Circuits (Inductor)

- ❖ An ideal inductor does not dissipate energy
 - It takes power from the circuit when storing energy
 - Delivers power to the circuit when returning previously stored energy
- ❖ A practical inductor has significant:
 - Resistance because of the
 - Conducting material
 - Capacitance because of the
 - Capacitive coupling between the conducting coils
- ❖ Impedance of an inductor is sL where
 - s is the complex frequency of the source signal
 - s for a **DC** signal is **0** and for a **sinusoidal AC** signal is $j\omega$
 - $\omega = 2\pi f$ where f is the frequency of the signal



Fundamentals of Electric Circuits (Inductor)

- ❖ Measured by
 - [Online Calculator](#)
 - Inductance meter
 - By [text or color marking](#) on the inductor
- ❖ Some useful links
 - [The Inductor](#)
 - [Inductance of a Coil](#)
 - [Mutual Inductance](#)
 - [Inductors in Series](#)
 - [Inductors in Parallel](#)

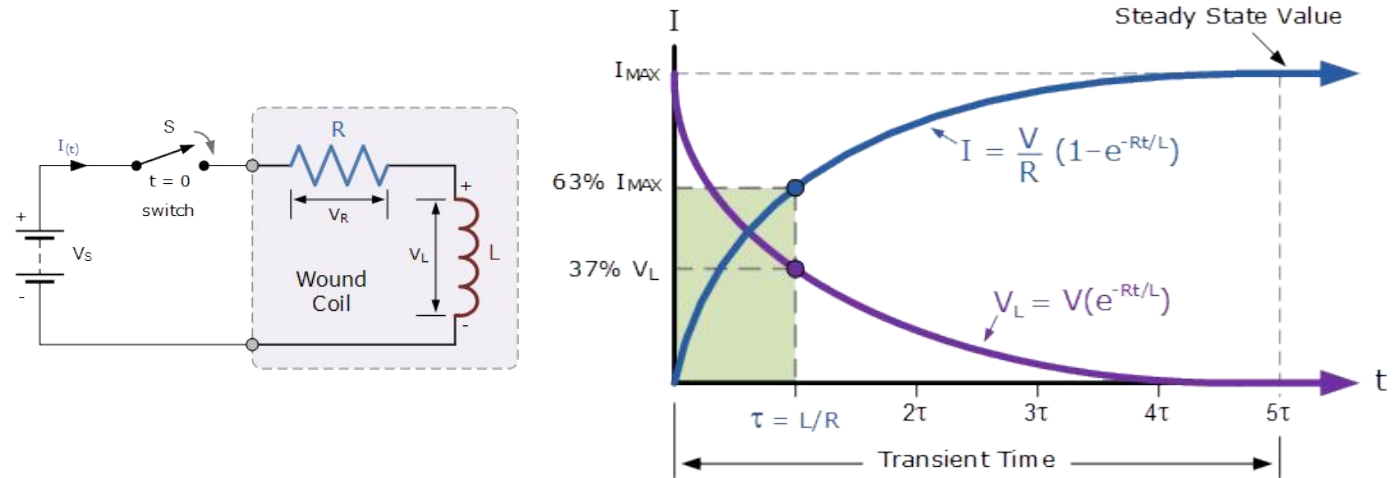


Various types of inductors: (a) solenoidal wound inductor, (b) toroidal inductor, (c) chip inductor.



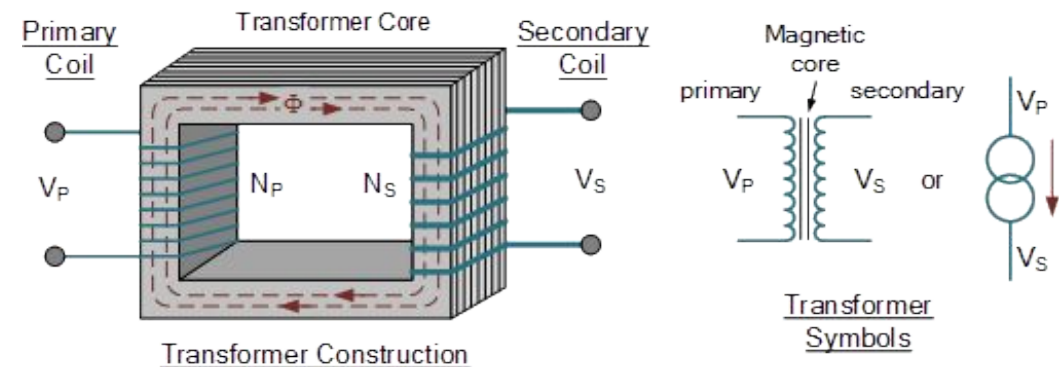
Fundamentals of Electric Circuits (Inductor)

- ❖ Store and release energy
 - [LR Series Circuit](#)
- ❖ AC, puls and etc. responses
- ❖ Applications
 - Filters
 - [Transformers](#)



$$\frac{N_P}{N_S} = \frac{V_P}{V_S} = n = \text{Turns Ratio}$$

$$\text{Power}_{\text{Primary}} = \text{Power}_{\text{Secondary}}$$

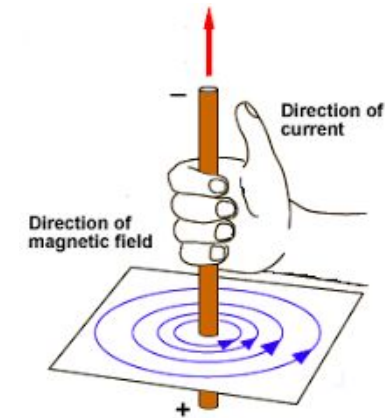
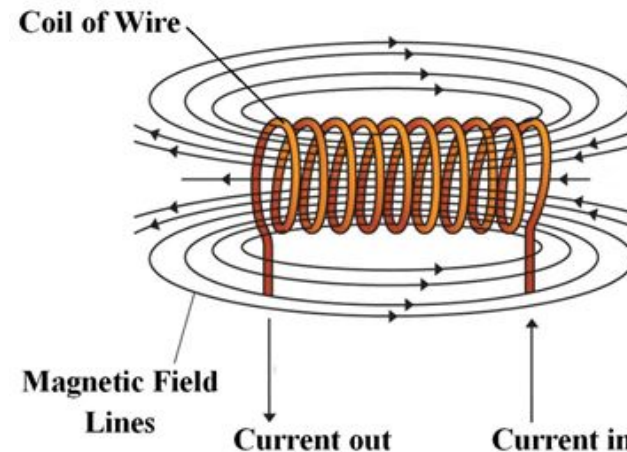
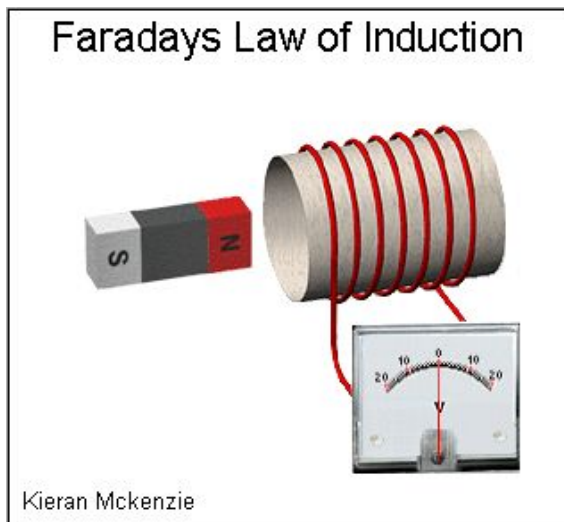


Fundamentals of Electric Circuits (Electromagnetism)

- ❖ A moving electric charge (**current**) generates a **magnetic field**
- ❖ A **magnetic field** induces electric charge movement (**current**)
- ❖ Electricity and magnetism are two related phenomena produced by the **electromagnetic** force
- ❖ In an [electromagnetic wave](#), the electric and magnetic fields are perpendicular to each other

[How Electric and Magnetic forces arise](#)

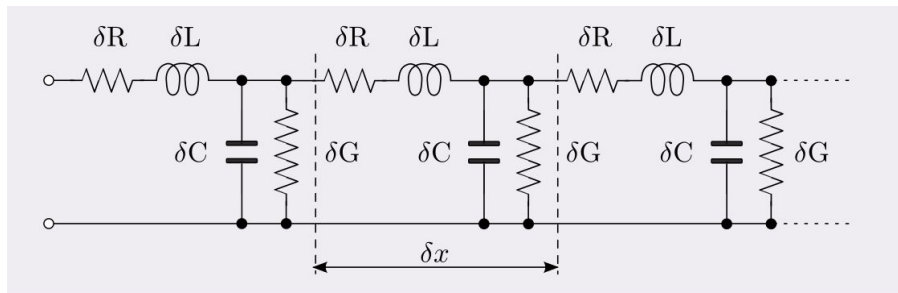
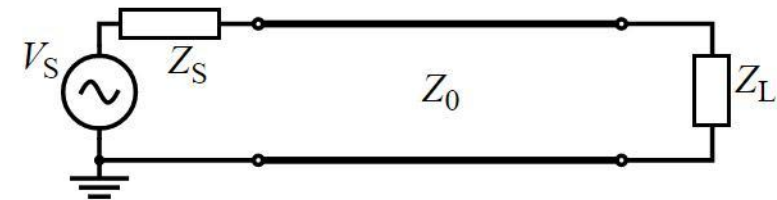
[The origin of Electromagnetic waves](#)



Fundamentals of Electric Circuits (Transmission Line)

- ❖ To deliver power, connection of two wires between the source and the load is required
- ❖ At low frequencies, power is delivered to the load in electric fields
- ❖ At higher frequencies, power is delivered in electric and magnetic fields
- ❖ Transmission line is a material structure that transfers energy in form of electromagnetic field
- ❖ If the impedances of the load and the transmission line are not matched
 - Some part of the energy will be reflected to the source
 - The reflection coefficient at the load is

$$\Gamma_L = \frac{Z_L - Z_o}{Z_L + Z_o}$$

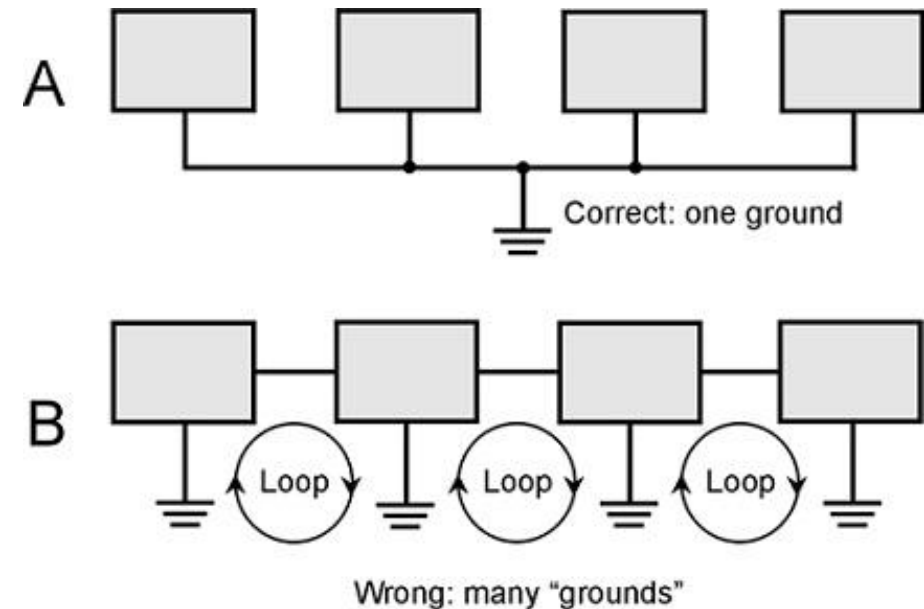


[Transmission Lines - Signal Transmission and Reflection](#)

Fundamentals of Electric Circuits (Ground Loops)

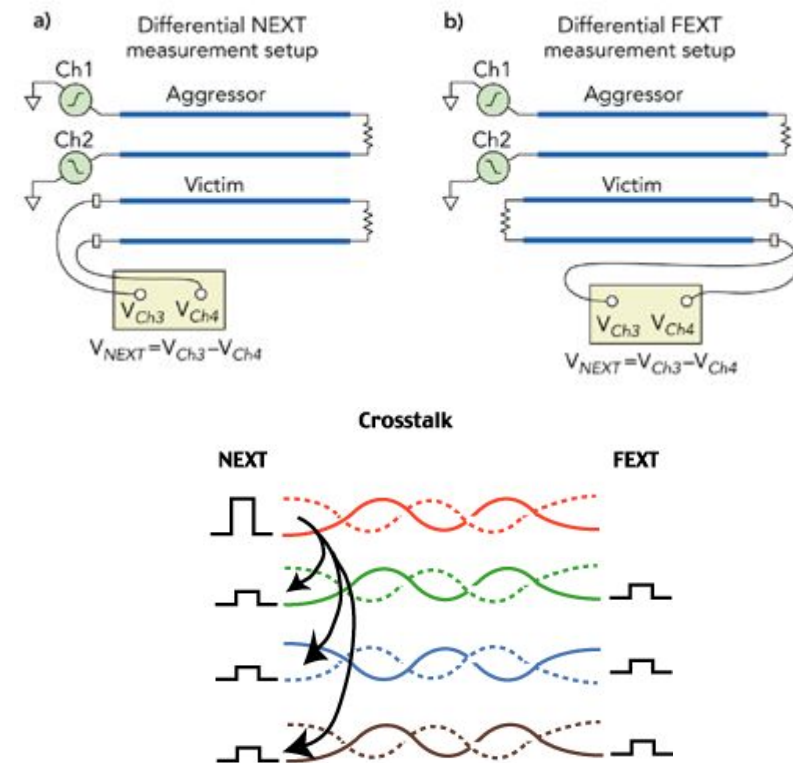
- ❖ Ground loop exists when an electrical system is connected through more than one path to the ground
- ❖ Ideally, all grounded points of an electrical system should be at the same potential
- ❖ The ground loop forms a large loop antenna that picks up interference currents easily
- ❖ The larger the loop the more interference (EMI)
- ❖ Result: The ground is no longer stable

- ❖ [Ground Loops: Avoid Them!](#)
- ❖ [Grounding and Shielding of electric circuits](#)
- ❖ [Galvanic Isolation: Purpose and Methodologies](#)



Fundamentals of Electric Circuits (Crosstalk)

- ❖ Electromagnetic (EM) Crosstalk is an unwanted interference
- ❖ It is caused by the electric and magnetic fields of one or more signals affecting another signal
- ❖ Crosstalk occurs via two mechanisms
 - Capacitive crosstalk caused by the electrical field
 - Inductive crosstalk caused by the magnetic field
- ❖ Some solutions
 - Shielding
 - Increase the distance between lines
 - Using twisted pair wires
- ❖ [Crosstalk \(electronics\)](#)
- ❖ [CrossTalk](#)



Fundamentals of Electric Circuits

❖ References

- Fundamentals of Electric Circuits, 5th Edition, Charles K. Alexander and Matthew N.O. Sadiku, ISBN 978-0-07-338057-5
- Fundamentals of Electrical Engineering, First Edition, Giorgio Rizzoni, ISBN 978-0-07-338037-7
- ***Most of the images have been copied from these two books***

❖ Some useful links

- [Decoupling Capacitors](#)
- [5 Lighting Terms You Should Know](#)
- [Impedance and Reactance](#)
- [Electronic Basics #12: Coils / Inductors \(Part 1\)](#)
- [Inductance basics](#)
- [Do Volts or Amps Kill You? Voltage, Current and Resistance](#)
- [How does an Antenna work? | ICT #4](#)
- [Electromagnetic interferences](#)
- [What is Hall Effect and How Hall Effect Sensors Work](#)

The SI prefixes.

Multiplier	Prefix	Symbol
10^{18}	exa	E
10^{15}	peta	P
10^{12}	tera	T
10^9	giga	G
10^6	mega	M
10^3	kilo	k
10^2	hecto	h
10	deka	da
10^{-1}	deci	d
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p
10^{-15}	femto	f
10^{-18}	atto	a